



Branko Zečević | 2026-04

PDF documents with Go (revisited)



Revisiting ?

Ref:

- GolangZg GoTalks 2024-06

What was then ?

- basics: what is PDF, how to create and view PDF ...
- common use: manuals, reports, contracts, invoices, form-based data entry, magazines, newspapers, books, document archiving, ...

What is today ?

- beyond basic editing and signing
- AI-(everything): document creation, content extraction, security, compliance, encryption, ...



AI-assisted PDF

AI-assisted software:

- intelligent text recognition, automated document editing, ...

Document automation:

- document content analysis, content extraction, content summary, automation of signing workflow, ...
- advanced encryption, digital signatures, watermarking, ...

Document creation:

- generate structured PDF from scratch, conversational interaction with PDF assistant software, ...

Collaboration and review:

- document review and summary, document version comparison, redaction of sensitive information, ...



Go and PDF

Status:

- not supported in standard library
- requires 3rd-party package w/ API to create and/or manipulate PDF documents
- diverse approaches

PDF processor, CLI and API support:

- pdfcpu, <https://github.com/pdfcpu/pdfcpu>

Document generator:

- folio, <https://github.com/carlos7ags/folio>
- maroto, <https://github.com/johnfercher/maroto>
- gofpdf, <https://github.com/jung-kurt/gofpdf> (archived!)
- unipdf, <https://github.com/unidoc/unipdf> (commercial!)

API-based service for conversion:

- gotenberg, <https://gotenberg.dev/>

Convert HTML to PDF:

- go-wkhtmltopdf, <https://github.com/SebastiaanKlippert/go-wkhtmltopdf>



Package:

- <https://maroto.io/>
- <https://github.com/johnfercher/maroto>
- MIT license

Features:

- grid system for layout positioning and alignment
- easy API
- component-based approach with elements (text, image, table, barcode, ...)
- fluent method interface (method calls can be chained together)
- support for styling (font, colour, size, ...)
- barcodes (EAN, code128, QR, DataMatrix)



Package:

- <https://github.com/carlos7ags/folio>
- <https://playground.foliopdf.dev/>
- Apache license

Features:

- layout engine
- HTML/CSS to PDF conversion
- text shaping for left-to-right, right-to-left, Indic and CJK scripts
- document redaction
- forms, digital signatures,
- barcodes,
- PDF/A compliance
- language SDKs: Go, Java, WASM



Folio-Maroto comparison

Feature	Folio	Maroto
layout engine	plan-based	grid-based
HTML to PDF	yes	no
i18n	yes: LTR, RTL, Indic, CJK	limited
forms	yes: AcroForms	no
digital signature	yes: PDF Advanced Electronic Signatures (PadES)	no
barcode	yes: Code128, EAN, QR	yes: Code 128, EAN, QR
redaction	yes	no
merge	yes	yes
standards	PDF/UA, PDF/A, auto-bookmarks, page labels	no
CLI	yes	no
language SDK	Go, Java, WASM	Go
dependency	Go x/ packages	Gofpdf



Plan-based layout ? Or grid-based ?

Plan-based layout:

- before rendering layout engine calculates "plan" for arrangement of elements on page: text, images, tables, ...
- plan includes: size and position of element, flow of elements across pages (page breaks), interaction of elements (for example: text wrapping around images, ...)
- layout is immutable: once created, plan cannot be changed, rendering process does not modify plan or elements, ...

Grid-based layout:

- content is placed within grid of rows and columns, where "cell" in grid can contain text, images, ...
- layout is defined by grid structure (similar to web layouts)
- automatic pagination: if content overflows page, new page is created automatically



Examples: Maroto

Sample program:

- for go 1.26.x, maroto v2

Tests:

- 1: single page, default configuration
- 2: single page, configuration a4 vertical, password
- 3: single page, configuration a4 vertical, margins
- 4: single page, configuration a4 vertical, header, footer, line
- 5: single page, configuration a4 vertical, header, footer, line, text, qr
- 6: multi page, configuration a4 vertical, custom font, header, footer, text, metrics
- 7: multi page, configuration a4 horizontal, fonts, header, footer, text
- 8: merge



Examples: Folio

Sample program:

- for go 1.26.x, folio v2

Tests:

- 1: single page, portrait
- 2: single page, portrait, security
- 3: single page, portrait, margins
- 4: multi page, portrait, margins, header, footer
- 5: single page, portrait, margins, header, footer, bookmarks
- 6: single page, portrait, margins, header, footer, html



References

PDF:

- <https://en.wikipedia.org/wiki/PDF>
- <https://www.iso.org/standard/75839.html>

PostScript:

- <https://en.wikipedia.org/wiki/PostScript>

Go software:

- <https://pkg.go.dev/>
- <https://golangweekly.com/>
- <https://awesome-go.com/>



Thank you!

Contacts:

Branko Zečević, branko.zecevic@pointer.hr

